

Solution Requirements

Date	9 July 2026
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely log in using username and password to access flood prediction services.	High
2	Authorization levels	Admin can manage datasets and predictions. Users can view flood predictions and receive alerts.	High
3	External interfaces	Integrates with Weather API, Rainfall Data, and Flood Dataset to generate accurate predictions.	High

		We build a web dashboard For citizens and admins report and view flood levels	High
4	Transactions processing	The system collects rainfall and weather data, processes it, predicts flood risk, and sends alerts to users.	High
5	Reporting	The system provides flood prediction reports, flood risk levels, and a dashboard for users and administrators.	Medium
6	Business rules, etc.	Flood alerts are generated only when the predicted flood risk exceeds the predefined threshold. Admin manages datasets, while users can only view predictions and alerts.	High
7	Compliance to laws or regulations	The system sends real-time flood alerts and provides flood prediction results through a user-friendly dashboard.	Medium
8	Other	The system sends real-time flood alerts and provides flood prediction results through a user-friendly dashboard.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	The system should predict flood risk and display results quickly. Prediction results within 3 seconds.	Prediction results within 3 seconds.
2	Scalability	The system should support many users and increasing flood data without performance issues.	Support 100+ users simultaneously.
3	Security & Data Privacy	User information and flood data should be securely stored and protected from unauthorized access	Secure login and encrypted data storage.
4	Reliability & Availability	The system should be available whenever users need flood predictions and alerts.	99% system availability.
5	Usability & Accessibility	The dashboard should be simple, user-friendly, and accessible on desktop and mobile devices.	Easy navigation with responsive interface.
6	<i>Other</i>	The system should allow future integration with weather APIs and real-time sensor data.	Easy integration with external data sources.